a Array-based ECS

Entity IDs map to array rows

Row	Position	Energy	Health
i	x[i]	e[i]	h[i]
i+1	x[i+1]	e[i+1]	h[i+1]
i+2	x[i+2]	e[i+2]	h[i+2]

Boolean activity / component masks

Sparse relationship matrices

Registered effect handlers

Batch updates to component arrays

Structural-command buffer

Entity and component changes

b Decisions, commit and event delivery

Client / SDK / server

sharp-421m

One-pass option scores

World Core

Validate / resolve

Apply registered effects

Batch

Intents

Brain

Bounded partitions

Pooled model clients

Atomic PostgreSQL transaction

World head/state + event outbox + command result

After commit

Outbox relay → NATS JetStream

Memory Context

Projection / query API

c Overlap across workload partitions

